
Computer Networks

IT351

Project

Deadline: Sunday 30/11/2025 @ 23:59

[Total Mark for this Project is 14]

Students Details:

CRN:

Name: Layan Khalid

ID: s230048771

Name: Jana

ID: s230032832

Abdulahakeem Miyajan

ID:

Name:

Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g., misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.

Learning Outcome(s):

Demonstrate protocol configuration, network addressing schemes, and analyze packet transmission for local and wide area networks.

Project

14 Marks

Introduction

Cisco Packet Tracer is an interactive simulation tool widely used for teaching and learning computer networking. It supports network design, configuration, and troubleshooting in Real-Time and Simulation modes. Through Packet Tracer, you can experiment with various networking protocols:

- **Layer 2:** Ethernet, PPP
- **Layer 3:** IP, ICMP, ARP
- **Layer 4:** TCP, UDP

It also supports routing techniques, enabling students to better understand real-world communication between devices. With its drag-and-drop interface, Packet Tracer allows users to build, test, and analyze networks efficiently.

Objective

This project is designed to help students:

- Gain familiarity with the Packet Tracer environment.
- Reconstruct a given network design and ensure it functions correctly.
- Practice IP addressing, routing, and connectivity testing across LANs.
- Apply troubleshooting skills to identify and fix communication issues.

Tools & Resources

You will need Cisco Packet Tracer installed on your computer. Download here: [Cisco NetAcad – Packet Tracer](#)

Tasks: (should be done by using packet Tracer)

1. Topology Setup: Build the network as shown in Figure 1.
2. Device Configuration: Assign IP addresses, subnet masks, default gateways, and DNS addresses to all end devices.
3. Switch Connections: Connect PCs to switches using the correct interfaces.
4. Router Connections: Connect switches to routers through the proper ports.
5. LAN Verification: Test communication within each LAN (LAN1, LAN2, LAN3).
6. Inter-LAN Connectivity: Ensure devices across LANs can communicate (e.g., LAN1 ↔ LAN2).

Learning Outcome(s):

Demonstrate protocol configuration, network addressing schemes, and analyze packet transmission for local and wide area networks.

Project

Network Topology

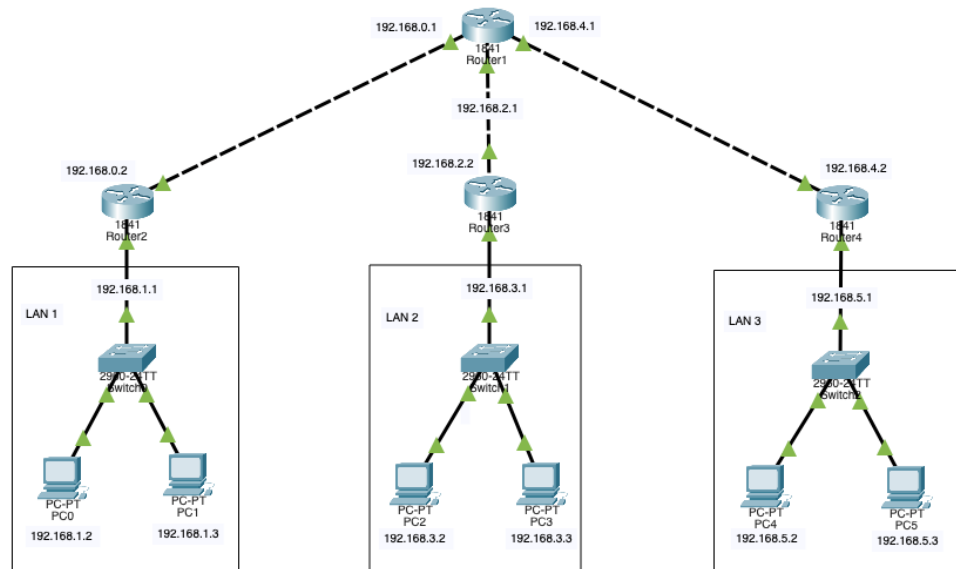


Figure 1: Topology of simple internetwork

Submission Guidelines

- Each group (3-4 members) must submit a single project.
- Clearly include group member names and CRN in the submission.
- For every task, provide the commands used and screenshots of the results. Missing steps or evidence will result in mark deductions.
- Submit the project on Blackboard before the deadline.

Marking Criteria (Total = 14 marks)

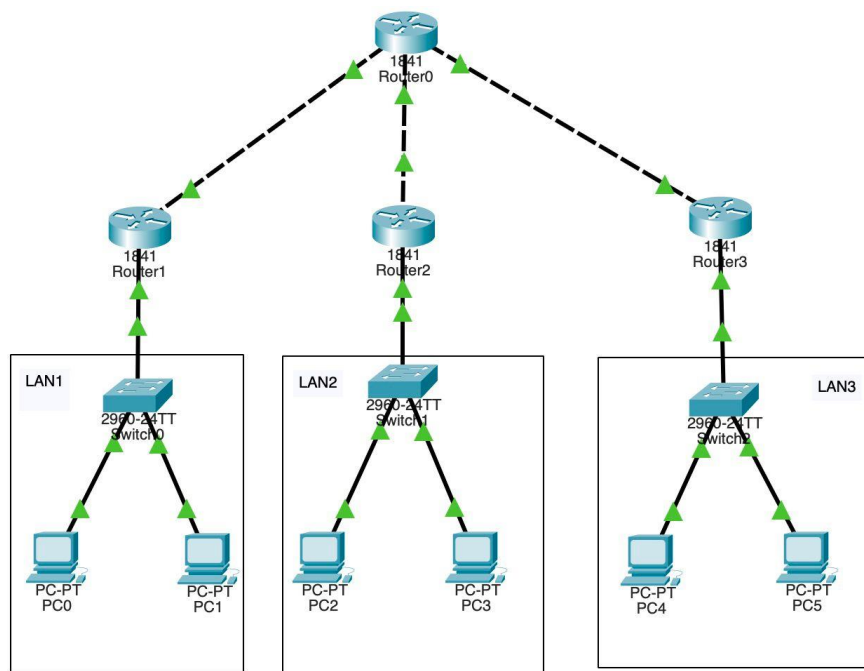
Marking Criteria	Marks
Creating topology as per the given diagram	3
Assigning Correct IP addresses	2
Connecting devices with switch through correct interfaces	2
Connecting switches with Router through correct interfaces	2
Making successful communication among the devices of LAN1	1
Making successful communication among the devices of LAN2	1
Making successful communication among the devices of LAN3	1
Making successful communication between two LANs	2
Total	14

Introduction

Cisco Packet Tracer is a simulation program used to develop and test computer networks.

It enables users to configure devices, assign IP addresses, and verify connectivity between different networks. This project helps students learn how data moves between networks and how to install routers and switches effectively.

1) Topology setup:



2) IP addresses:

-Device1 : PC0....

The screenshot shows the configuration window for PC0, specifically the 'Desktop' tab. The 'IP Configuration' section is active, showing settings for the 'FastEthernet0' interface. The 'Static' option is selected for both IPv4 and IPv6 configurations. The IPv4 settings are: IP Address 192.168.1.2, Subnet Mask 255.255.255.0, Default Gateway 192.168.1.1, and DNS Server 192.168.1.1. The IPv6 settings are: Static selected, Link Local Address FE80::2D0:D3FF:FE51:E506, and empty fields for IPv6 Address, Default Gateway, and DNS Server. The '802.1X' section shows 'Use 802.1X Security' unchecked, 'Authentication' set to 'MD5', and empty fields for 'Username' and 'Password'. A 'Top' button is at the bottom left.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.1
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	/
Link Local Address	FE80::2D0:D3FF:FE51:E506
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

☐ Top

-Device2: PC1....

PC1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.1.3

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

192.168.1.1

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::2D0:97FF:FEAE:946B

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

-Device3: PC2.....

PC2

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.3.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.3.1

DNS Server

192.168.3.1

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::201:97FF:FE6A:9656

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

-Device4: PC3....

PC3

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address192.168.3.3

Subnet Mask255.255.255.0

Default Gateway192.168.3.1

DNS Server192.168.3.1

IPv6 Configuration

Automatic

Static

IPv6 Address /

Link Local AddressFE80::2E0:A3FF:FED0:E43

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

-Device5: PC4...

PC4

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address192.168.5.2

Subnet Mask255.255.255.0

Default Gateway192.168.5.1

DNS Server192.168.5.1

IPv6 Configuration

Automatic

Static

IPv6 Address /

Link Local AddressFE80::230:A3FF:FE1B:73D5

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

مقيّد - Restricted

-Device6: PC5...

PC5

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.5.3

Subnet Mask

255.255.255.0

Default Gateway

192.168.5.1

DNS Server

192.168.5.1

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::2D0:FFFF:FEE5:9C8A

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

-Router 0...

Router0

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Ethernet0/1/0

Ethernet0/1/0

Port Status

On

Link Speed

10 Mbps

Auto

Duplex

Half Duplex

Full Duplex

Auto

MAC Address

00E0.A393.9446

IP Configuration

IPv4 Address

192.168.4.1

Subnet Mask

255.255.255.0

Tx Ring Limit

10

Equivalent IOS Commands

```
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Ethernet0/1/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Ethernet0/1/0
Router(config-if)#
```

Top

-Router 1...

Router1

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

FastEthernet0/1

Port Status☒ On

Link Speed

☐ 100 Mbps☐ 10 Mbps☒ Auto

Duplex

☐ Half Duplex☒ Full Duplex☒ Auto

MAC Address0001.425A.8802

IP Configuration

IPv4 Address192.168.1.1

Subnet Mask255.255.255.0

Tx Ring Limit10

Equivalent IOS Commands

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#

☐ Top

-Router 2...

Router2

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

FastEthernet0/1

Port Status

On

Link Speed

100 Mbps10 MbpsAuto

Duplex

Half DuplexFull DuplexAuto

MAC Address

0060.3E8E.C302

IP Configuration

IPv4 Address

192.168.3.1

Subnet Mask

255.255.255.0

Tx Ring Limit

10

Equivalent IOS Commands

Press RETURN to get started!

Press RETURN to get started!

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#

Top

-Router 3...

Router3

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

FastEthernet0/1

Port Status

On

Link Speed

100 Mbps10 MbpsAuto

Duplex

Half DuplexFull DuplexAuto

MAC Address

00D0.BC57.8B02

IP Configuration

IPv4 Address

192.168.5.1

Subnet Mask

255.255.255.0

Tx Ring Limit

10

Equivalent IOS Commands

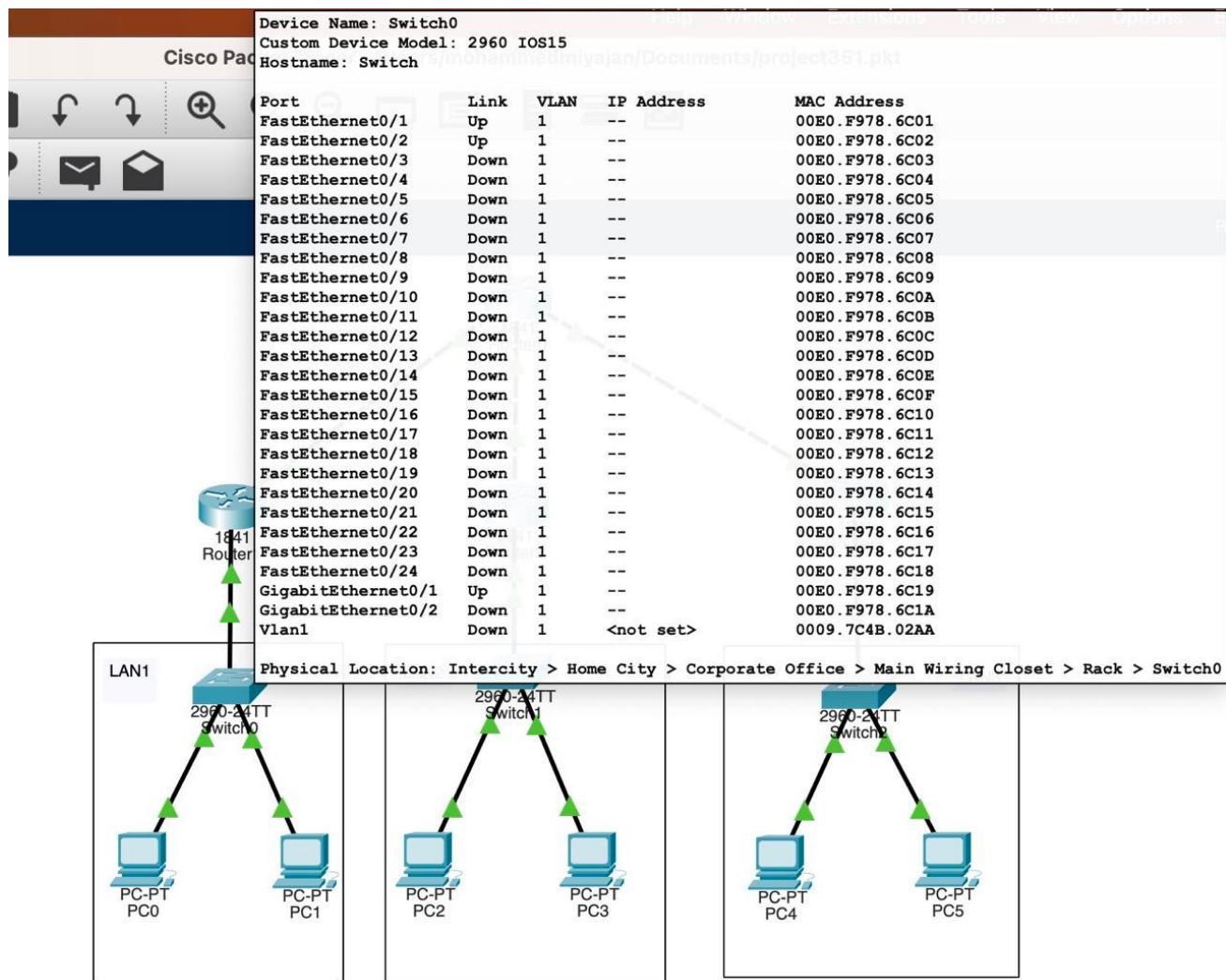
Press RETURN to get started!

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#

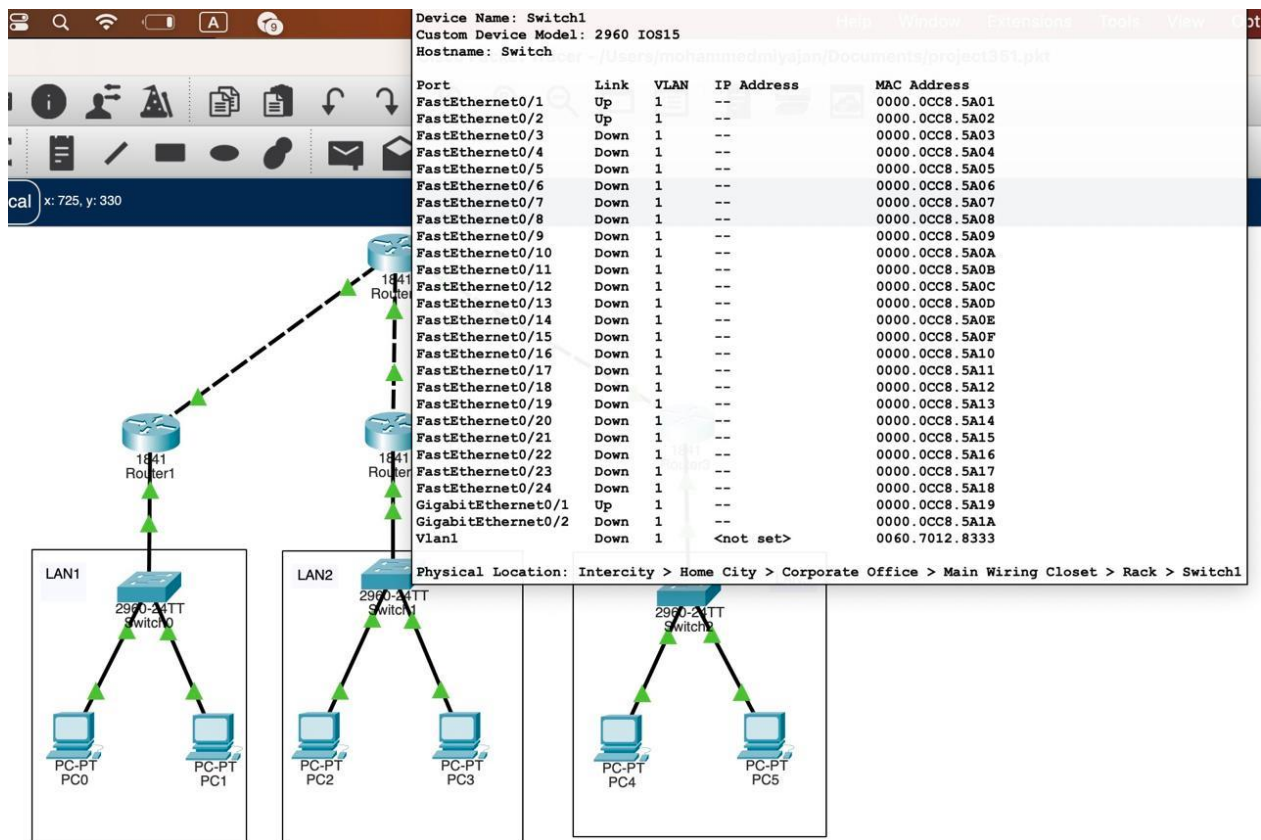
Top

3) Switch Connections:

■ Switch0:



■ Switch1:



■ Switch 2:

Device Name: Switch2
Custom Device Model: 2960 IOS15
Hostname: Switch

Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	1	--	0007.EC4C.5101
FastEthernet0/2	Up	1	--	0007.EC4C.5102
FastEthernet0/3	Down	1	--	0007.EC4C.5103
FastEthernet0/4	Down	1	--	0007.EC4C.5104
FastEthernet0/5	Down	1	--	0007.EC4C.5105
FastEthernet0/6	Down	1	--	0007.EC4C.5106
FastEthernet0/7	Down	1	--	0007.EC4C.5107
FastEthernet0/8	Down	1	--	0007.EC4C.5108
FastEthernet0/9	Down	1	--	0007.EC4C.5109
FastEthernet0/10	Down	1	--	0007.EC4C.510A
FastEthernet0/11	Down	1	--	0007.EC4C.510B
FastEthernet0/12	Down	1	--	0007.EC4C.510C
FastEthernet0/13	Down	1	--	0007.EC4C.510D
FastEthernet0/14	Down	1	--	0007.EC4C.510E
FastEthernet0/15	Down	1	--	0007.EC4C.510F
FastEthernet0/16	Down	1	--	0007.EC4C.5110
FastEthernet0/17	Down	1	--	0007.EC4C.5111
FastEthernet0/18	Down	1	--	0007.EC4C.5112
FastEthernet0/19	Down	1	--	0007.EC4C.5113
FastEthernet0/20	Down	1	--	0007.EC4C.5114
FastEthernet0/21	Down	1	--	0007.EC4C.5115
FastEthernet0/22	Down	1	--	0007.EC4C.5116
FastEthernet0/23	Down	1	--	0007.EC4C.5117
FastEthernet0/24	Down	1	--	0007.EC4C.5118
GigabitEthernet0/1	Up	1	--	0007.EC4C.5119
GigabitEthernet0/2	Down	1	--	0007.EC4C.511A
Vlan1	Down	1	<not set>	0060.70CB.8758

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Switch2

LAN3

4) Router Connections:

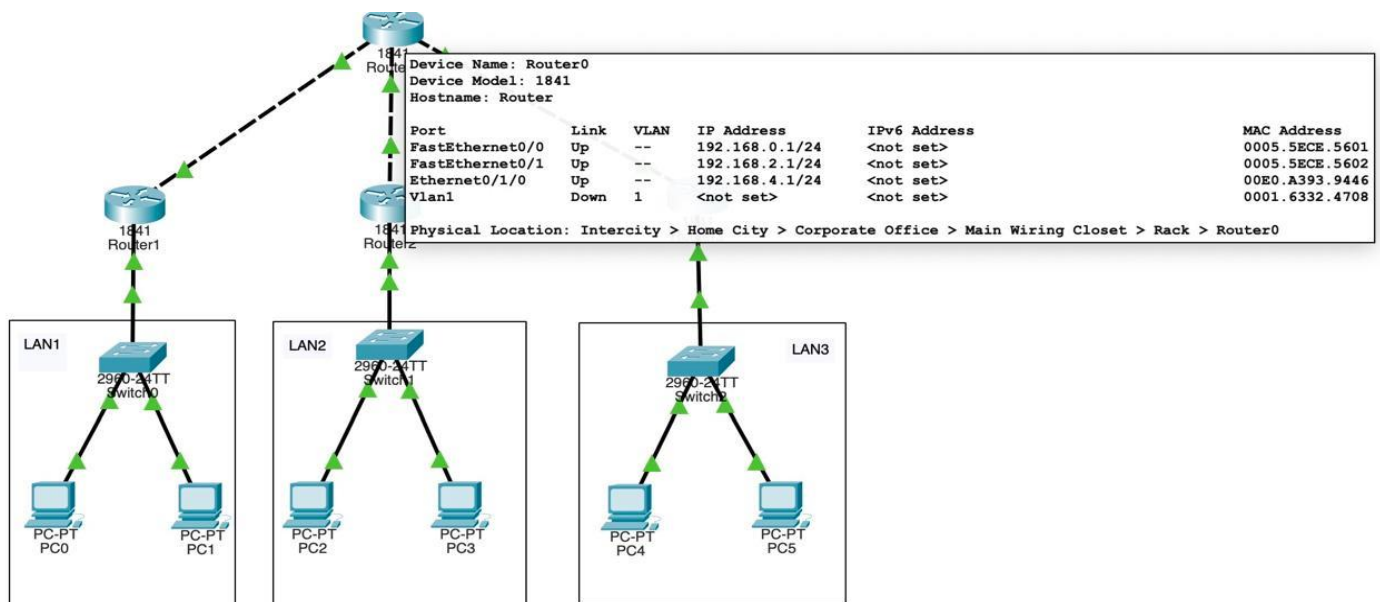
■ Router0...

To access peripheral local area networks (LAN1, LAN2, LAN3):

```
Router0(config)#ip route 192.168.1.0 255.255.255.0 192.168.0.2
```

```
Router0(config)#ip route 192.168.3.0 255.255.255.0 192.168.2.2
```

```
Router0(config)#ip route 192.168.5.0 255.255.255.0 192.168.4.2
```



■ Router1...

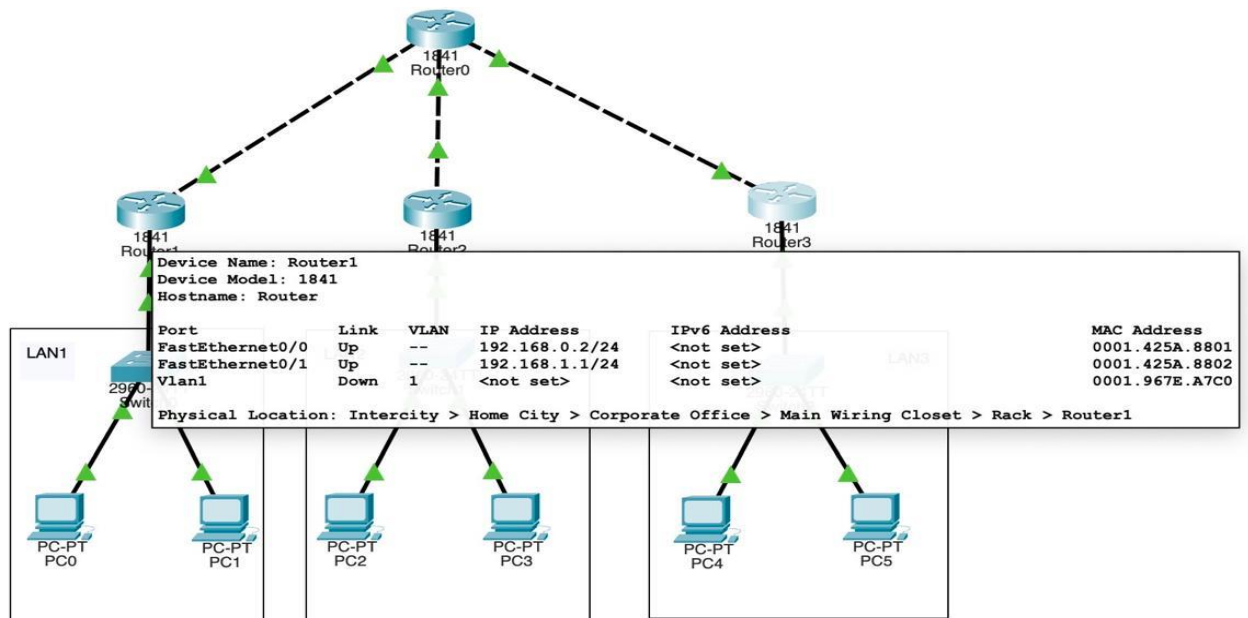
To access remote networks (LAN2, LAN3 and other interconnected networks) via Router0 (Next jump):

Router1(config)#ip route 192.168.2.0 255.255.255.0 192.168.0.1

Router1(config)#ip route 192.168.3.0 255.255.255.0 192.168.0.1

Router1(config)#ip route 192.168.4.0 255.255.255.0 192.168.0.1

Router1(config)#ip route 192.168.5.0 255.255.255.0 192.168.0.1



■ Router2...

To access remote networks (LAN1, LAN3 and other interconnected networks) via Router0 (next hop:

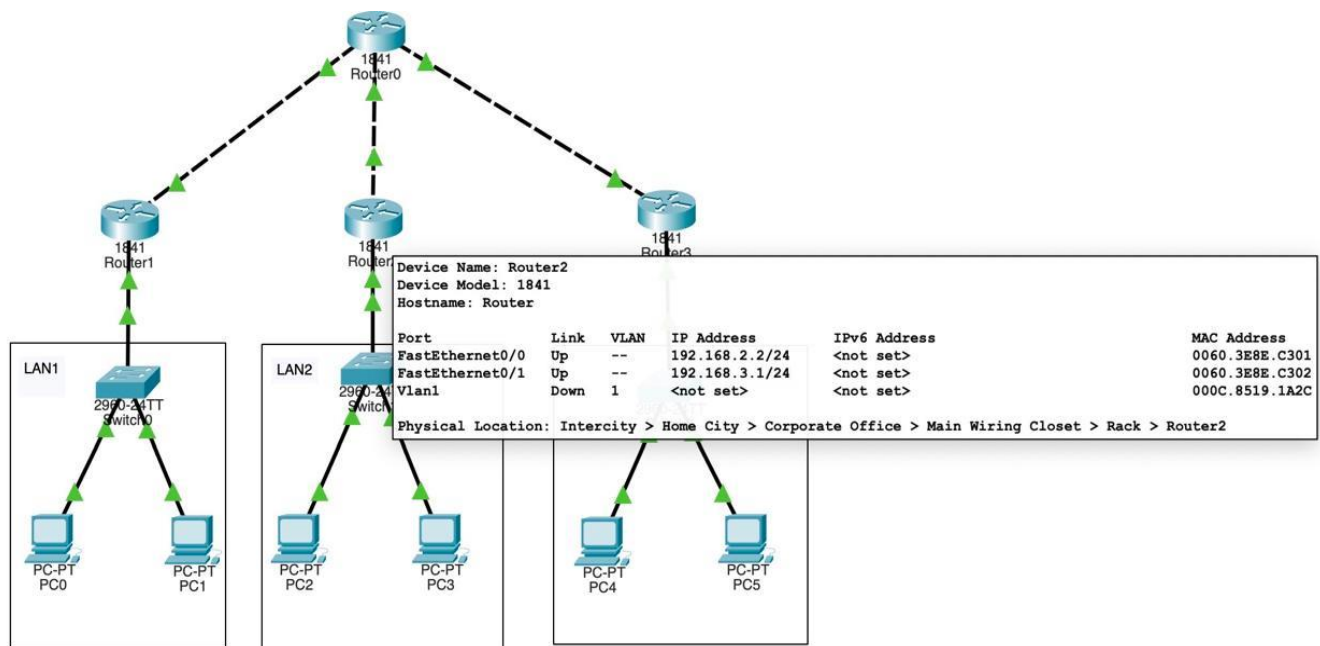
192.168.2.1):

Router2(config)#ip route 192.168.0.0 255.255.255.0 192.168.2.1

Router2(config)#ip route 192.168.1.0 255.255.255.0 192.168.2.1

Router2(config)#ip route 192.168.4.0 255.255.255.0 192.168.2.1

Router2(config)#ip route 192.168.5.0 255.255.255.0 192.168.2.1



■ Router3...

To access remote networks (LAN1, LAN2 and other interconnected networks) via Router0 (next hop:

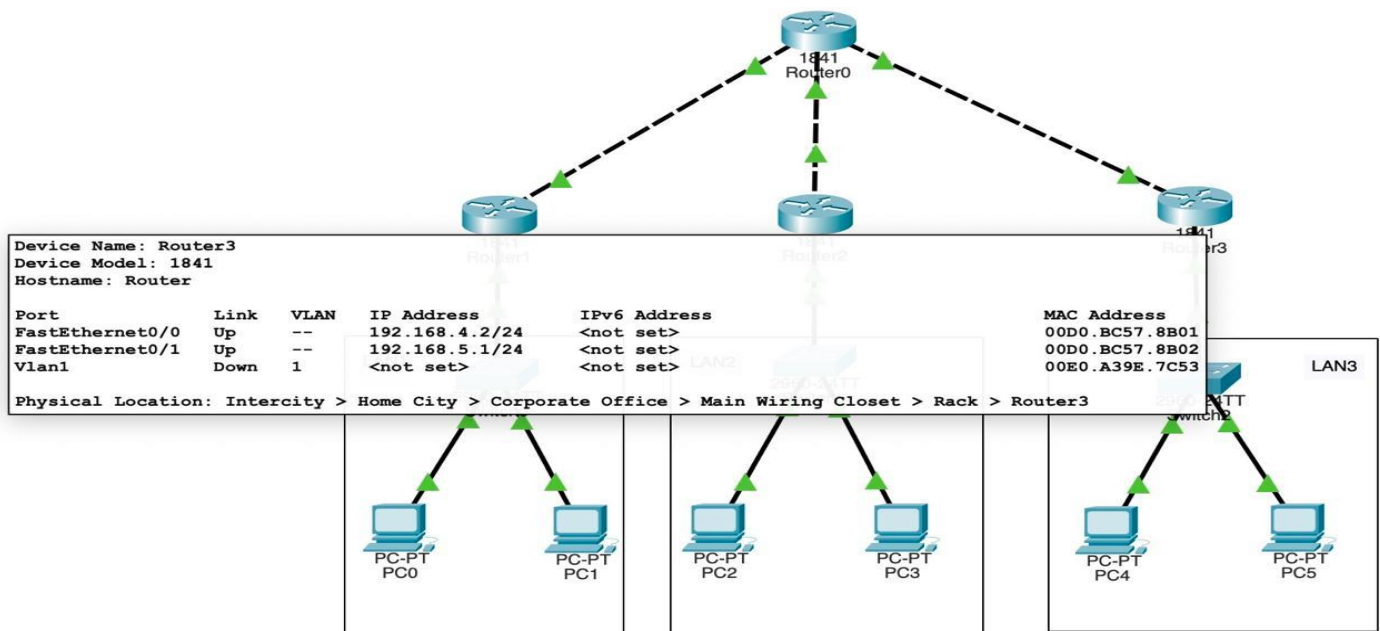
192.168.4.1):

Router3(config)#ip route 192.168.0.0 255.255.255.0 192.168.4.1

Router3(config)#ip route 192.168.1.0 255.255.255.0 192.168.4.1

Router3(config)#ip route 192.168.2.0 255.255.255.0 192.168.4.1

Router3(config)#ip route 192.168.3.0 255.255.255.0 192.168.4.1



5) LANs Verification:

- For successful communication among the devices of LAN1:

```
C:\>ping 192.168.1.3

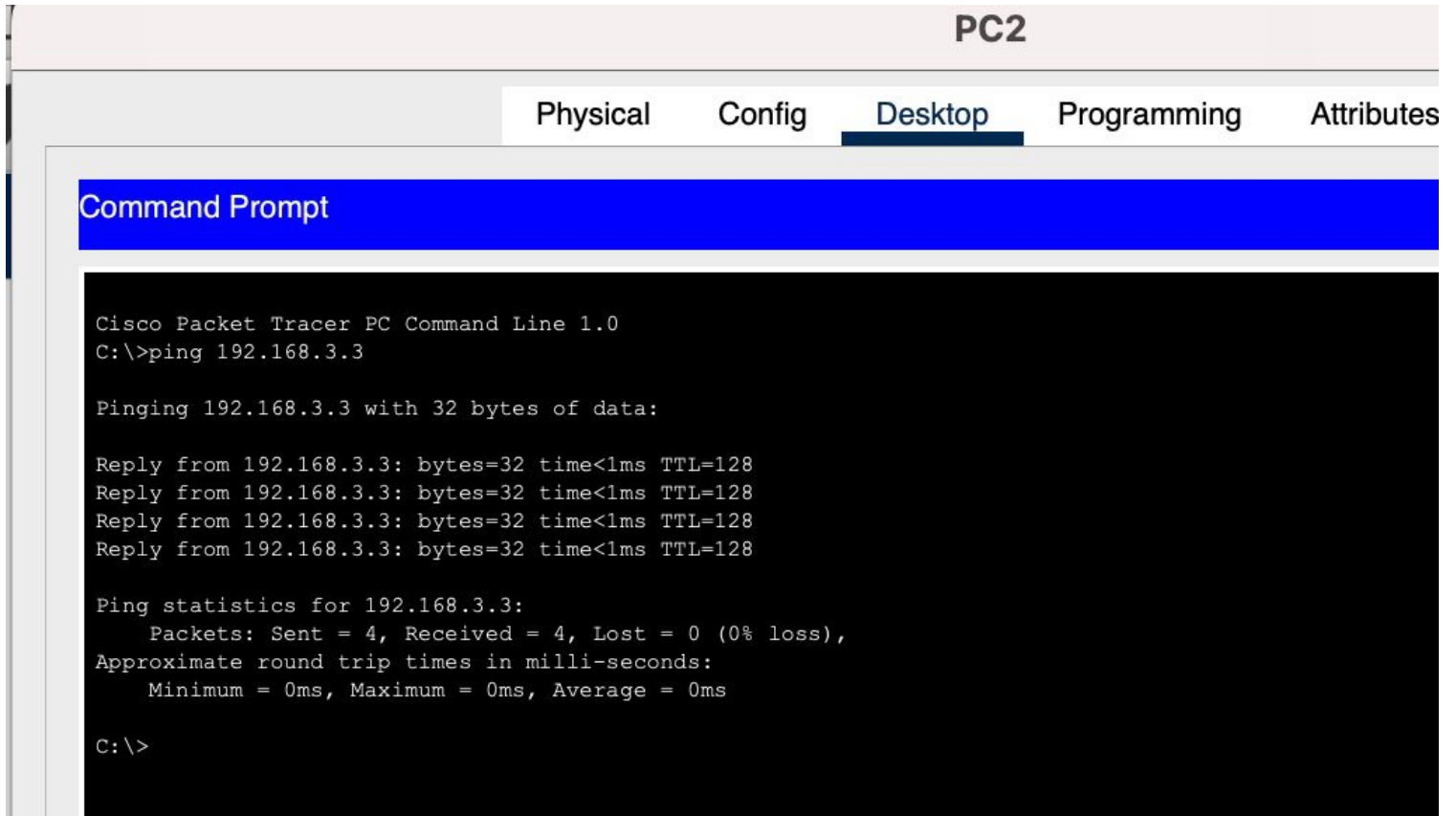
Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=15ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

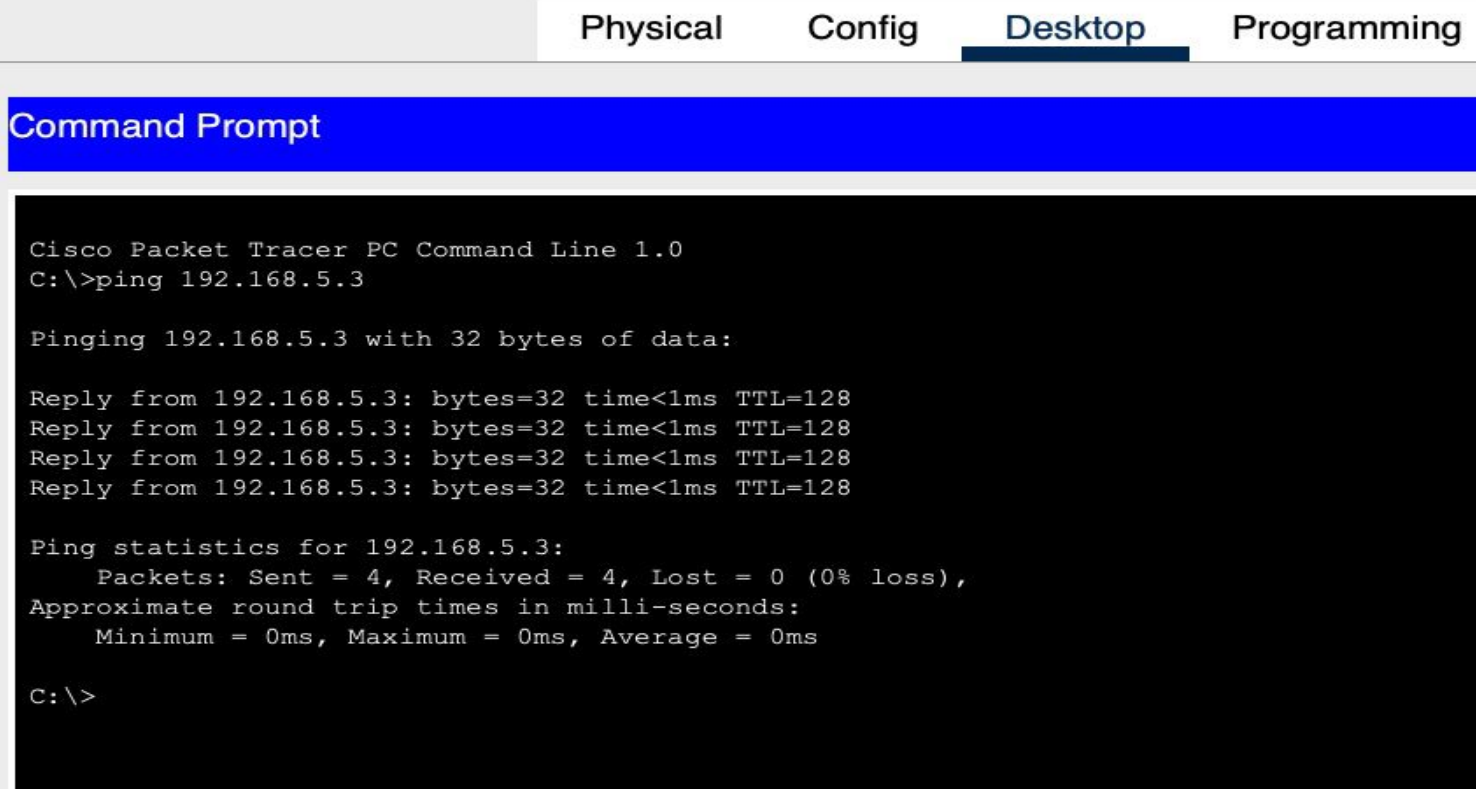
Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>|
```

- For successful communication among the devices of LAN2:



- For successful communication between devices of LAN3:



The image shows a screenshot of the Cisco Packet Tracer Desktop tab. At the top, there are four tabs: Physical, Config, Desktop (which is selected and highlighted in blue), and Programming. Below the tabs is a blue header bar with the text "Command Prompt". The main area of the window is black with white text, displaying the output of a ping command. The text shows the command "C:\>ping 192.168.5.3" being entered, followed by the response "Pinging 192.168.5.3 with 32 bytes of data:". This is followed by four lines of "Reply from 192.168.5.3: bytes=32 time<1ms TTL=128". Below this, the "Ping statistics for 192.168.5.3:" are shown, indicating "Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)", and "Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms". The prompt "C:\>" is visible at the bottom.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.5.3

Pinging 192.168.5.3 with 32 bytes of data:

Reply from 192.168.5.3: bytes=32 time<1ms TTL=128
Reply from 192.168.5.3: bytes=32 time<1ms TTL=128
Reply from 192.168.5.3: bytes=32 time<1ms TTL=128
Reply from 192.168.5.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.5.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

6) Inter LANs-connectivity:

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:D3FF:FE51:E506
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 192.163.5.3

Pinging 192.163.5.3 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Request timed out.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.163.5.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Conclusion:

We learned how to set up and test a network using Cisco Packet Tracer thanks to this assignment.

We discovered how IP addressing, subnetting, and routing facilitate network-to-network communication; additionally, it enhanced my configuration and troubleshooting abilities.

